

Reminder: Empirical and simulation exercises must be submitted with “stand-alone” code and data (if any) in an appropriate format. The homework is due Sunday (data, code, and word-processed report by email to **the instructor before 10 PM Sunday**). Also, submit your word-processed report via **Gradescope**.

1. Download US quarterly real GDP (seasonally adjusted). Construct the quarterly growth rate of real GDP by taking log differences. Subtract the sample mean to focus on the autocovariance. This question asks you to estimate the population spectrum of the GDP growth rate using various methods:

- Plot the periodogram divided by 2π . The periodogram $I(\omega_k)$ is known to be $|f_k|^2$. Also, notice that `fft` in Matlab returns f_k in the lecture slides with a different normalization. Specifically, `fft` in Matlab returns $\sum_{t=0}^{T-1} x_t \exp(-it\omega_k)$ without the scaling constant $\frac{1}{\sqrt{T}}$ (HINT: $\frac{|f_k|^2}{2\pi} = \text{abs}(f_k).^2 / 2 / \pi$ in Matlab).
- Fit an AR(p) model, where p is chosen by AIC. You can use OLS to estimate the parameters. Plot the spectrum of the estimated model.
- Estimate the spectrum using the Newey-West kernel and the sample autocovariance.
- Add a pre-whitening step to c. Use AR(1) for the pre-whitening step.

Discuss the differences and similarities. Repeat a similar investigation for GDP deflator inflation rates and federal funds rates. Do the same analysis with the HP filtered cyclical components. You may want to use `hpfiler` in Matlab.

2. Kuznets noted a long swing in economic activity with a periodicity of approximately 20 years. To make his conclusion, he did the following. Suppose that we have an annual series x_t . Then,

Step1: take 5 years centered moving average: $y_t = \frac{1}{5}(L^{-2} + L^{-2} + 1 + L + L^2)x_t$.

Step 2: take a centered difference in the averaged data: $z_t = y_{t+5} - y_{t-5} = (L^{-5} - L^5)y_t$.

Let $K(L)$ be the filter above, i.e., $z_t = K(L)x_t$. Plot the gain of the Kuznets filter $|K(e^{-i\omega})|$. Where does it have a peak? Do you think this filter may generate a long swing when applied to a white noise process?

3. Consider two independent series,

$$\begin{aligned}x_t &= x_{t-1} + e_t, & e_t &\sim WN(0,1), \\y_t &= y_{t-1} + u_t, & u_t &\sim WN(0,1),\end{aligned}$$

where $\{e_t\} \perp \{u_t\}$. Suppose we obtain cyclical variations in x and y using the HP filter. The cyclical components are denoted by x_t^c and y_t^c . We want to see the distribution of $\text{corr}(x_t^c, y_t^c)$.

- Simulate x_t and y_t . The sample period is $T = 120$, corresponding to 30 years at the quarterly frequency (HINT: `normrnd` and `cumsum` in Matlab).
- Apply the HP filter.
- Compute the correlation between the filtered series (`corr` in Matlab).
- Repeat a-c. 2,000 times.
- Plot the histogram of the simulated correlations (`hist` in Matlab).

Discuss the risk of obtaining artificial comovements (or cross-correlations) between x and y for a sample of reasonable sizes, although x and y are not related in the population. Repeat the same exercise for the band-pass filter of Baxter and King. Use $K = 12$. If you want, you may use the program written by others, e.g., <https://ideas.repec.org/c/wpa/wuwprr/0507001.html>.

4. Suppose you have a quarterly series x_t . The cyclical component obtained by applying the HP filter, c_t , is denoted by $H(L)x_t$. In a lecture, we saw that

$$H(L) = \frac{\lambda(1-L)^2(1-L^{-1})^2}{1 + \lambda(1-L)^2(1-L^{-1})^2}.$$

Now, consider a non-stationary process x_t . Specifically, we assume that x_t is a pure random walk:

$$x_t = x_{t-1} + e_t, \quad e_t \sim WN(0, \sigma^2).$$

In this case, we can understand the HP filter as a two-step filter. First, it applies $(1-L)$ to derive a stationary process with a well-defined autocovariance structure. That is, we define $y_t = (1-L)x_t$, which is e_t in our example. Then, the second step is to apply the remaining component of the filter, $H^+(L) = \frac{\lambda(1-L)(1-L^{-1})^2}{1 + \lambda(1-L)^2(1-L^{-1})^2}$, to y_t . Note that the initial series x_t does not have specific dynamics at business cycle frequencies (it is based on white noise). This question is designed to investigate the dynamics of the filtered series c_t .

- Plot the gain of H^+ . Will $\{c_t\}$ have more variation at the business cycle frequencies than others?
- Show that the spectrum of the filtered series $\{c_t\}$ is given by $s_c(\omega) = \frac{\sigma^2}{2\pi} |H^+(e^{-i\omega})|^2$ for $\omega \in [-\pi, \pi]$.
- Show that $\int_{-\pi}^{\pi} s_c(\omega) e^{ik\omega} d\omega = \gamma_c(k)$, where $\gamma_c(k) = E[c_t c_{t-k}]$ for $k \in Z$.
- Suppose that $\sigma = 1$. Compute the autocovariances of $\{c_t\}$ at lags 0, 1, ..., 10 by numerically evaluating $\int_{-\pi}^{\pi} s_c(\omega) e^{ik\omega} d\omega$. To do so, first, set a equi-spaced grid $\{\omega_1, \dots, \omega_N\}$, e.g., `linspace(-pi, pi, N)` in Matlab for a reasonably large N . Second, approximate $d\omega$ by $\omega_2 - \omega_1$. Third, compute $\{s_c(\omega_1), \dots, s_c(\omega_N)\}$. Finally, for each k , calculate

$$\sum_{l=1}^{N-1} s_c(\omega_l) e^{ik\omega_l} (\omega_2 - \omega_1),$$

as an approximation to $\int_{-\pi}^{\pi} s_c(\omega) e^{ik\omega} d\omega$. Plot the result.